

Tutorial: Spaceplane basics

If you're tired of big rockets that use tons of fuel and disintegrate in the atmosphere when coming back to Kerbin, this tutorial is perfect for you! This tutorial will help you with the basics of spaceplane flight, and will help you avoid the most common errors that could ruin your day as a spaceplane pilot! This is just a general briefing section with lots of "to do" or "not to do" things: when you think you've got it, check the [Aeris 4A tutorial mission](#) to learn how to get into space easily. Note: This tutorial was last updated for version 1.7.2.

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Jet section

While having the same stock parts as in the VAB, spaceplane design is obviously quite different from pure rocket design. First of all, since the launch happens horizontally, you will have to include [landing gears](#), and you will most likely want to include [jet engines](#) for the first stage for excellent fuel and cost efficiency.

Engines

- *Main article:* [Jet engine](#)
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While it's true that jet engines don't work in space, they offer one large advantage over rocket engines while inside the atmosphere: fuel efficiency. One FL-T100 tank can't power any rocket into space, yet a Shock Cone Intake, a Mk1 Inline Cockpit, a half-filled FL-T100 and a J-X4 "Whiplash" Turbo Ramjet Engine aimed in the general direction of "up" will let you laugh your way past the 70km mark at nearly 1200m/s. The Whiplash's ridiculous fuel efficiency allows a spaceplane to climb high into the atmosphere and gain a lot of speed while barely using any fuel at all.

Rapiers generally provide less thrust than a Whiplash at speeds below mach 2, but provide more thrust at higher speeds. They are able to operate at higher altitudes and can even continue operating in a vacuum by switching to a rocket mode. This makes them a very effective choice for SSTOs since they do a good job of getting maximum velocity out of air-breathing mode, and then allow you to use oxidizer without needing the added weight and drag of a separate rocket engine.

LV-N "Nerv" nuclear thermal rockets are commonly used on SSTOs for the rocket stage if Rapiers are not used. These have an ISP of 800 in a vacuum which is significantly better than all other rocket motors except the very weak IX-6315 ion rocket. LV-N engines also have the advantage of using the same liquid for propellant that jet engines use for fuel. This makes design easier, eliminating all concern for balancing jet fuel against rocket fuel. However, they are extremely heavy for their power, weighing as much as a conventional rocket nearly 11 times more powerful. This is all right if their high efficiency saves enough fuel, but that may not be the case in small spaceplanes with limited fuel capacity.

Note: Some high-efficiency rocket engines lose most of their power and efficiency in low atmosphere. LV-N has less than 25% of its full power at Kerbin sea level. However, it's not a matter of "atmosphere or not", just a matter of air pressure which decreases rapidly with altitude. LV-N exceeds 75% of its full power at just 7700m altitude on Kerbin.

T-1 "Dart" engines are unique amongst conventionally-fuelled rocket engines: They have close to the highest efficiency in both vacuum and low atmosphere. This is good for spaceplanes operating in low-oxygen atmospheres where jet engines can't be used. However, they do take a conventional mix of fuel and oxidizer, and thus require much more fuel mass than jet engines or LV-N Nerv.

Air intakes

→ *Main article: Air intake*

Obviously jet engines are air-breathing, so you need to include air intakes in your spaceplane. Note that most air intakes tend to produce less drag than aerodynamic nose cones and pointy cockpits, especially shock cone intakes. As such, you may want to line all forward Mk1 radial nodes on your aircraft with shock cones, if possible. Also note that shock cone intakes appear to provide the best air supply at all mach speeds, and unlike other air intakes, they do not decline in performance beyond certain mach speeds. (Source (<http://forum.kerbalspaceprogram.com/index.php?/topic/125015-105-intakes-lets-figure-them-out/>))

Be warned that if you do not provide sufficient air intakes for the engines you've placed, you may find that some of your engines shut down before others. In an aircraft with two or more engines, this can potentially cause you to enter a flat spin which can be unrecoverable if your center of mass

is behind your center of lift. However, don't bother going overboard with intakes because your engines will still shut off at high altitudes due to the low air pressure regardless of how many intakes you have.

Fuselages

Temperature tolerance is the primary consideration for fuselage choice. The Mk1 fuselage parts only tolerate up to 2000K, while Mk2 fuselages tolerate up to 2500K, and Mk3 fuselages tolerate up to 2700K. Consequently, atmospheric reentry is less dangerous with a Mk2 or Mk3 fuselage, compared to reentry with Mk1 fuselages. Note that no wings will tolerate more than 2400K, so the difference in temperature tolerance between Mk2 and Mk3 fuselages isn't terribly significant. You probably won't have much luck landing the fuselage intact if your Mk3 plane gets its wings scorched off on reentry.

One final point to consider is the mass you're planning to store in the fuselage. More mass will mean it takes more drag to slow down on reentry, which will mean you go faster at lower altitudes and experience more reentry heating. If your Mk3 design jam-packed full of heavy gear can't seem to survive reentry, one option might be to reduce the payload a bit.

Fuel storage

Whether you're storing your fuel in fuselage sections, wing sections, or attached inside of cargo bays, it's generally a good idea to keep equal amounts of fuel at equal distances from your center of mass. This helps to keep the performance of your spaceplane stable with any amount of fuel. If you keep all of the fuel in the front, you may find that your center of mass drifts backwards as your fuel drains. This can turn into a fatal scenario if the center of mass gets behind the center of lift and you enter a flat spin. Conversely, if you keep all of the fuel in the back, you may find that your center of mass gradually drifts so far in front of your center of lift that you can't keep your nose up anymore, also potentially resulting in a fatal scenario. Balanced fuel saves Kerbal lives.

Cargo

If you plan on either dropping off cargo or picking up cargo and traveling with it, you'll generally want to locate your cargo bay at the middle of your center of gravity. This is for the same reason that you keep your fuel balanced.

Note that dropping off cargo from the back with a Mk3 Cargo Ramp is bad for more than one reason. While all other cargo bays are fine for making spaceplanes, the Mk3 Cargo Ramp produces obscene amounts of drag, which can easily prevent reaching orbital velocities by itself. A relatively low-drag alternative is to use an inverted cargo bay and some hydraulic cylinders with structural panels as a cargo elevator.

Wings

→ *Main article:* [Wing](#)

As you would expect, spaceplanes need wings: they have various shapes and dimensions, and they differ basically in lift rating: you will want to have enough lift to keep your fuselage approximately prograde during your ascent to orbit. Wing shapes do not appear to have a significant impact on

drag or drag-to-lift ratio at this time, but swept wings are still an easy way to help ensure that your center of lift stays behind your center of mass. Note that the lift rating does not mean that the wings will automatically lift a spaceplane into the air when it's moving forward. All lift-rating means is that the wing section will resist motion perpendicular to its plane. To takeoff and land at low speed, it's helpful to rotate the wings so the leading edge is slightly above the trailing edge. Ideally, the wings should be tilted upward at an angle about $3\text{--}5^\circ$ from the fuselage for optimal lift-to-drag ratio.^[1]

For spaceplanes, avoid the FAT parts (wing, tail fin, and control surface). These have extremely poor temperature tolerance and will almost inevitably break up during atmospheric reentry. Also avoid the basic fin for the same reason.

If you need to carry more fuel, consider using the Big-S Delta Wing as it provides the same lift-to-drag and lift-to-weight ratio as any other wing of the same mass, yet has the ability to carry fuel as well. This allows you to avoid the extra weight and mass of additional fuselage parts, so you only have to deal with the extra weight of the fuel itself.

Control surfaces

→ *Main article:* [Control surface](#)

It is also advisable to add some control surfaces to your plane to have some control in the atmosphere: you can manually add them to the wings or choose winglets with effective control surfaces, like the Standard Canard. Keep in mind that lift rating and control surfaces are not connected: lift rating is basically the capacity of your wings to sustain the weight of your spaceplane, while control surfaces are parts of wing that can be moved to change the flow of the air around the plane and through this change a plane's direction, angle of attack or inclination. This is an important distinction; a plane with great lift rating but without any control surfaces will fly easily but will be almost uncontrollable. Vice versa a plane with lots of control surfaces will be perfectly controllable (maybe even too much) but may have big difficulties with taking off and landing at reasonable speeds.

Control surfaces are heavier than wings. A control surface with 100% control surface portion will weigh twice as much as a wing with the same lift would weigh. As such, don't use control surfaces as substitutes for wings, and don't use more control surfaces than you need.

It is advised to place your control surfaces as far from your center of mass as possible. Ailerons to roll your aircraft should be placed as far off to the edges of your wings as possible. Canards and horizontal tail fins should be placed as far towards the front and back of your aircraft as possible, respectively. To minimize drag, vertical tail fins can be kept minimal, since aircraft can be maneuvered adequately by roll and pitch alone. Note that canards are somewhat more efficient than horizontal tail fins since canards provide an upward force with upward rotation, and downward force with downward rotation.

Also note that for maximum efficiency, you should make sure that your horizontal control surfaces are rotated to exactly the same pitch that you've rotated your wings.

Landing gears

→ *Main article:* [Landing gear](#)

It can be helpful to use a slightly taller front landing gear and slightly shorter rear landing gears so that your fuselage points slightly upward on the ground, thereby increasing the angle of attack of your wings while on the ground. This allows you to takeoff at lower speeds and on shorter runways, and likewise for landing. To maximize lift, your aircraft should rest on the ground with the fuselage tilted upward at anywhere up to a 25-27° angle so that the wings will end up tilted back at up to 30°. Lift maxes out at 30° and declines beyond that point, so avoid excessive fuselage tilt.^[1]

It's strongly recommended to keep your landing gears well-spaced from each other to ensure that the aircraft will be difficult to roll into a collision. For example, having your landing gears located near the ends of the wings is an easy way to ensure that you don't roll and shred your wings when landing or taking off. Having landing gears near the front and back of your aircraft can also help to ensure that you won't break your engines or smash your cockpit into the runway.

Safety note: Disable the brakes on the front landing gear. Remember how you want your center of lift/drag to be behind the center of gravity? The same principle applies here. If the front landing gear touches down first and has its brakes on, it's going to make the rest of your aircraft pivot around it as it's pressed into the ground. This can easily cause you to crash on landing. Brakes in the back keep you stable.

If you can maintain level flight at about 30-40 m/s, you should be able to perform an ocean landing if needed. If you're planning on landing on a somewhat uneven surface, like an open grassland somewhere on Kerbin or an island on Laythe, consider packing some parachutes for deceleration. You can deploy your chutes just prior to touchdown for rapid deceleration.

Piloting

Spaceplanes generally ascend best on a 10-30 degree incline, often leveling out towards higher altitudes to pick up the maximum horizontal velocity before switching to rocket motors. Some testing is usually required with new designs to determine the best ascent profile.

Note that as you fly higher the air intakes will become less effective and you may come to a point when the engines will shut down due to the lack of air. At that point, the plane could potentially start spinning around as a result of losing the benefit of the gimbal control of the engine. This would indicate two problems. Firstly, inadequate air intakes for the number and type of engines you're using (causing some engines to shut down before others). Secondly, it would suggest that your spaceplane's center of lift is too far forward compared to its center of gravity (causing the uncontrollable spin). You can resolve the emergency by transferring fuel from rear fuel tanks to forward fuel tanks, but you should alter your design to bring your wings further back to prevent such incidents in the future.

If, instead, all goes well, then once your jet engines have shut down you will need to start the second part of your space mission by switching to rockets. Note that when your jet engines shut down simultaneously while climbing in otherwise normal flight, it is generally due to a lack of pressure from the altitude you're flying at. Increasing the number of intakes will not allow you to continue using your jet engines at higher altitudes.

Rocket section

When your jet engines stop working, it is time to ignite the old, reliable liquid fuel engine. Keep in mind that as your altitude increases, your control surfaces and winglets will become increasingly ineffective and will no longer work at all once you leave the atmosphere, so you may need to add alternative control systems like reaction wheels or RCS thrusters.

As long as you're in space, your spaceplane won't differ from any spacecraft: you will probably want to add batteries and generators to prevent the command pod from running out of power. It is also common to add an Inline Clamp-O-Tron, which, unlike all the other docking ports, can be placed in the middle of the spacecraft (a handy solution, since there is not much space at either end of the craft) to allow your plane to dock with space stations or other spacecrafts.

Deorbiting

The most dangerous part of a spaceplane flight is returning from orbit. Re-entry heating can destroy parts of your spaceplane, or destroy it entirely. After that, you face the challenge of touching down on the ground and coming to a stop safely without rolling and breaking off a wing or taking a nosedive and blowing your aircraft up on touchdown, or rolling off of the runway and into the ocean.

Atmospheric re-entry

To survive re-entry, it's recommended to start your approach back into the atmosphere at a shallow angle, ideally with a periapsis of around 30-35 km. Keep your spaceplane pointed about 90 degrees above prograde so that the wings and body of your aircraft slow you down as much as possible. You may also want to deploy your landing gears to increase your drag, as well as airbrakes if you have them. As you approach 35-50 km, your aircraft will most likely level itself out, at which point you can try aiming about five degrees above the horizon line. If your aircraft is burning up during this stage, you may need larger wings to slow you down faster, radiator panels to carry away the heat more effectively, parts with a higher temperature tolerance (like the Mk2 liquid fuel fuselage instead of the Mk1 liquid fuel fuselage), or parts to increase your maneuverability, like RCS thrusters, reaction wheels, or canards and elevons.

Alternatively, if you're returning from a high orbit or from an interplanetary trip, you can try repeated shallow passes through the atmosphere. With initial passes, you start off with a periapsis at 50-60 km and gradually lower your apoapsis until you're in an almost circularized low orbit and then transition to your usual re-entry approach. If you've been able to successfully re-enter on previous low orbit test runs, you should be able to use this method to achieve similar rates of success once you've slowed down sufficiently.

Landing

Landing can be trickier for spaceplanes since they are designed for higher speeds than other aircraft and may not be able to fly level at speeds low enough for an easy landing (<50 m/s). When dealing with high-speed landings, you may touch down too quickly and cause the front of the plane to smack into the runway. After a successful touchdown, high-speed motion on the runway (let alone uneven ground) can be unstable, causing the aircraft to careen to one side or the other, potentially resulting in loss of a wing and sometimes the entire aircraft. Landing also often requires rapid deceleration to avoid running off the end of the runway or crashing into a slope when landing on open terrain.

All of these problems can be exacerbated or reduced by adjusting the amount of fuel in your tanks during landing. Close to empty tanks will allow you to fly slower, decelerate faster, and reduce touchdown strain. Ideally, you ought to test landing the spaceplane with full fuel tanks and with nearly empty fuel tanks prior to taking your spaceplane to orbit. If your spaceplane is difficult to land when full of fuel but easy to land when empty, then it may be helpful to burn off or transfer out most of the excess fuel before landing to make the aircraft lighter. In subsequent missions, you may want to launch with less fuel to cut down on cost and make landing a quicker and easier process.

To recover the most value from your spaceplane, you should try to land on the runway at the Space Center ([this tutorial](#), although it has been written for spacecraft and not spaceplanes, is a great help). If that's not an option, you can still recover some value by landing at any suitable flat place on Kerbin. You can even try refueling it before recovering your spaceplane further increasing your recovering value. Be aware that landing on water is an option if your spaceplane can fly level at less than ~40-50 m/s.

Landing speed

Landing speed (minimum speed for level flight) can be reduced by adding components to increase maneuverability, by using larger wings, by increasing wing angle of attack on the fuselage (3-5 degrees is the recommended range for a spaceplane to achieve the best lift-to-drag ratio [source (<http://forum.kerbalspaceprogram.com/index.php?/topic/124380-wing-lift-amp-wing-lift-to-drag-ratio-charts/>)]), and by decreasing the weight of the aircraft. When landing, you can achieve the highest lift for a given speed by raising the total angle of attack of your wings to 30 degrees (although this induces a great deal of drag). Be aware that while this angle will provide the lowest possible landing speed for your spaceplane, it will create problems if your center of gravity is too far ahead of your rear landing gears (causing a high-speed nose collision on touchdown) or if your rear landing gears are too far ahead of your engines, causing them to strike the runway first.

If your spaceplane is able to fly and land steadily at low speeds but just you're having difficulty slowing down as you approach the runway, try to reach your desired speed first and then approach the runway in almost level flight. You can slow down either by deploying landing gears (and airbrakes if you have them) or by repeatedly pitching up and then back down to increase your drag.

If you have trouble pitching up enough to land at a reasonable speed, you can increase your maneuverability by toggling your flaps, canards and ailerons to greater than 100% control authority.

Touchdown collisions

These occur at their worst when your center of gravity is far ahead of your rear landing gears and you have a heavy plane at high speeds and a high angle of attack on landing, resulting in your front landing gear rapidly striking the runway after your rear landing gears touchdown. Consider placing your rear landing gears close to the spaceplane's center of mass, but be careful to avoid an engine collision with the runway. It helps to have a center of gravity which is close to your engines so that

the landing gears can be close to both. Alternatively, you can try landing at higher speeds with your nose pointed further down, but this increases challenges with stability and deceleration while on the runway.

Landing stability

When your spaceplane rolls shortly after touchdown, veers to one side and then explodes on the runway, you have a problem with landing stability. This is generally an issue of not spacing your landing gears out far enough apart. For an example, see the A-10 Warthog's landing gears: [link \(https://upload.wikimedia.org/wikipedia/commons/5/59/Thunderbolt.a10.fairford.arp.jpg\)](https://upload.wikimedia.org/wikipedia/commons/5/59/Thunderbolt.a10.fairford.arp.jpg). Notice how the landing gears are placed out on the wings. For a Mk1-based aircraft, your rear landing gears should not be tightly tucked together on the fuselage. Mk2 aircraft may be able to get away with it, but even then it's often best to space them further out. However, make sure to use struts when placing landing gears on the far edges of a multi-part wing because they may sag enough to cause a fuselage collision with the runway during landing.

Make sure that all of your landing gears are pointing in exactly the same direction. Even a small deviation can cause serious instability, making your aircraft bounce and jolt left and right even during takeoff.

To avoid swerving on takeoff and landing, it is strongly recommended to turn off or reduce the strength of the front brake on your aircraft, as well as to reduce the friction control. An active front brake can cause your aircraft to rapidly and uncontrollably pivot left or right during landing, and high friction from the front wheel can potentially cause your plane to swerve violently both in landing and takeoff. The only drawback to the reduced friction is reduced steering control, so this setting may need to be adjusted when taxiing to/from a runway, but should otherwise be kept minimal for takeoff and landing.

Deceleration

Even with a stable landing, you may find that you don't have enough room on your desired landing area to come to a stop before you reach the end. To minimize the risk of such a situation, try to land on a large patch of flat open ground approaching a downhill slope. A good example of this is at the KSC runway when landing on a 90 degree bearing. If you can't slow down in time, you can simply flick your engines back on to take off and turn around for another try as you pass over the coast. In contrast, if you attempt a landing at the KSC runway on a 270 degree bearing, you run the risk of colliding with the upward slope shortly beyond the runway if you can't slow down initially and then can't speed up fast enough. Similar principles apply when finding suitable landing sites away from the KSC.

To slow down faster, you can increase the braking strength of your rear wheels. You can also use parachutes on landing, but care must be taken to ensure that an adequate length of runway remains since you'll only get one chance to use them. If you have a very short length of runway remaining and your parachutes can't slow you down fast enough, you'll be forced to cut the chutes and attempt a second landing without them. That said, parachutes are an exceedingly effective means of reducing your stopping distance. Note that a Wheesley or Goliath engine can reverse its

thrust to allow rapid deceleration during landing, but these are not recommended for a spaceplane due to being unreasonably heavy and inducing excessive drag when attempting to transition to orbital velocities.

See also

- [Tutorial: Basic Plane Design](#)
- [Tutorial:Basic SSTO Design](#)
- [Tutorial:Spaceplane to Laythe](#)

References

1. <https://forum.kerbalspaceprogram.com/index.php?/topic/124380-wing-lift-amp-wing-lift-to-drag-ratio-charts/>

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